Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CS5DS04]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

MACHINE LEARNING

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. What is Machine Learning? Explain different perspective and issues in machine learning. **07**

   b. Differentiate between supervised, unsupervised and semi supervised training. Explain with suitable examples. **07**

2. a. A dataset has the value of **07**

   X= [0.00, 0.22, 0.24]

   Y= [0.00, 0.22, 0.58]

   The value of θ_0(slope), θ_1 is 0.45 and 0.75, respectively. Y is representing the target dataset.

   Calculate the mean square error (MSE) and sum square error (SSE).

   b. Explain about Linear Discriminant Analysis with an example. **07**

3. a. For the following set of training samples, find which attribute can be chosen as the root for decision tree classification and justify your answer with example. **07**

Instance	Classification	A1	A2
1	+	T	T
2	+	T	T
3	-	T	F
4	+	F	F
5	-	F	T
6	-	F	T

b.

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

Predict that if "Red, SUV, and Domestic" will come together, can the car be stolen or not by using Naïve Bayes?

OR

4. a. What is a decision tree & discuss the use of decision tree for classification purpose with an example. **07**

b. Find out the information gain of overlook for the given data. **07**

		Play Golf	
		Yes	No
Overlook	Sunny	1	2
	Overcast	2	3
	Rainy	3	4

5. a. This is a 2D dataset **07**

A= {(1,2), (1,4)}

B= {(2,3), (2,4)}

Which technique will be useful to dimensionality reduction under unsupervised learning? Apply dimensionality reduction technique on this dataset and minimize the dimension.

b. Explain the Normalization and Standardization and in which condition it should use as preprocessing in machine learning model? **07**

OR

6. a. This is a 2D dataset **07**

A= {(1,2), (1,4)}

B= {(7,10), (6,8)}

Choose a suitable technique for the dimensionality reduction under supervised learning? Apply dimensionality reduction technique on this dataset and minimize the dimension. Examine the dataset before and after dimension reduction and explain weather, dimension reduction is better for this dataset or not to draw the decision boundary.

b. 1. A system is getting 99% and 49% accuracy during training and testing time, respectively. Examine the type of problem system is facing and choose a suitable solution for this type of issues? **07**

2. A system is getting 49% and 47% accuracy during training and testing time, respectively. Examine the type of problem system is facing and choose a suitable solution for this type of issues?

7. a. Determine the equation of a straight line which best fits the following data **07**

Year	2000	2001	2002	2003	2004
Sales	35	36	79	80	40

b. The sales of a commodity in tones varied from January 2010 to December 2010 as follows: **07**

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sales	280	240	270	300	280	290	210	200	230	200	230	210

Fit a trend line by the method of semi-average.

OR

8. a. Explain with example the following **07**

1. Time Series Patterns
2. Trend Pattern
3. Seasonal Pattern

b. Explain Linear Trend Regression with suitable example **07**